



Evaluation of the chemopreventive effects of Ankaferd Bloodstopper in 7,12-dimethylbenz[a]anthracene-induced oral epithelial dysplasia

Murat Ozle¹ · Dilek A. Uğar Çankal¹ · Mert İlhan^{2,3} · Hikmet Keleş⁴ · Esra Küpeli Akkol²

Received: 5 April 2016 / Accepted: 22 February 2018 / Published online: 2 March 2018
© Springer-Verlag GmbH Germany, part of Springer Nature 2018

Abstract

Objectives Ankaferd BloodStopper® (ABS) is an herbal extract which has been used historically as a hemostatic agent in traditional Turkish medicine. ABS comprises of standardized mixture of herbs *Thymus vulgaris*, *Glycyrrhiza glabra*, *Vitis vinifera*, *Alpinia officinarum*, and *Urtica dioica*. In addition to its hemostatic effects, the herb ABS contains some other biological effects including antioxidant and antitumoral properties. The aim of this study is to investigate the chemopreventive effects of ABS in 7,12-dimethylbenz[a]anthracene (DMBA)-induced oral epithelial dysplasia.

Materials and methods A total of 40 Sprague Dawley rats were randomly divided into four groups. Group 1 animals received DMBA alone, and group 2 animals received both DMBA and Ankaferd. Group 3 animals received ABS alone while group 4 animals served as control group and received only liquid paraffine. All animals were sacrificed, and tissue samples were analyzed histologically at the end of the experimental period (14 weeks).

Results Histological studies have shown that the buccal pouches of animals treated with DMBA alone revealed severe dysplasia while only mild or no dysplasia were noticed in DMBA + ABS group. Ankaferd were administered to animals and control group showed no dysplasia or other oral lesions.

Conclusion The results suggest that Ankaferd Bloodstopper® has chemopreventive effect against DMBA-induced oral epithelial dysplasia.

Clinical relevance Ankaferd Bloodstopper® could be used as a supportive treatment option of cancer in oral and maxillofacial surgery since it possesses chemopreventive effect.

Keywords Ankaferd BloodStopper® · Antioxidant · 7,12-Dimethylbenz[a]anthracene · DMBA · Dysplasia

Introduction

Human death causes change due to the development in technology and culture. Due to increasing industrialization, exposure to the chemical carcinogens has been on the rise [1]. Thus, cancer is a growing health issue and is

a leading cause of death worldwide, accounting for 8.2 million deaths in 2012 [2]. Oral and pharyngeal cancer is an important component of the worldwide burden of cancer. Oral cancer incidence has been rising over the last few decades [3]. An estimated 263,900 new cases and 128,000 deaths from oral cavity cancer (including lip cancer) occurred in 2008 worldwide. More than 50,000 new oral cavity and pharynx cancer patients are estimated in 2008 in the USA [4, 5]. However, survival rates for oral cancer have not yet shown significant improvement over the past 50 years and are among the lowest of major cancers. The 5-year and 10-year relative survival rates are 59 and 48%, respectively [3, 6]. Worldwide, carcinomas of the head and neck region account for more than 5% of all malignancies, and 90% of head and neck region carcinomas are squamous cell carcinomas (SCC) [6, 7].

SCC usually begins as an epithelial dysplasia at the stratified squamous epithelium. Dysplastic cells exceed

✉ Esra Küpeli Akkol
esrak@gazi.edu.tr

¹ Department of Oral and Maxillofacial Surgery, Faculty of Dentistry, Gazi University, 06510 Ankara, Turkey

² Department of Pharmacognosy, Faculty of Pharmacy, Gazi University, 06330 Ankara, Turkey

³ Department of Pharmacognosy, Faculty of Pharmacy, Yüzüncü Yıl University, 65080 Van, Turkey

⁴ Department of Pathology, Faculty of Veterinary Medicine, Afyon Kocatepe University, 03200 Afyonkarahisar, Turkey

basal layer and extends to connective tissue [8]. Under the widely used World Health Organization (WHO) classification for the pathological diagnosis of oral premalignant lesions, dysplasia, which is graded as mild, moderate, or severe, and carcinoma in situ (CIS) are classified as precursor lesions of oral squamous cell carcinoma, and if untreated, oral epithelial dysplasia should transform into SCC [9, 10].

7,12-Dimethylbenz[a]anthracene is an immune suppressor agent and an organ-specific carcinogen [11]. DMBA has been widely used in cancer studies over 50 years [12, 13]. DMBA is commonly used to create oral, skin, and breast cancer models. DMBA-induced oral carcinogenesis in animal studies is preceded by a sequence of precancerous lesions, which are very similar to that of human oral cancerous lesions, histopathologically, morphologically, and at molecular level [11].

Ankaferd Bloodstopper® (ABS) is a topical hemostatic agent which can be used effectively in the clinical practices such as control of external bleedings, traumatic injuries, dental and periodontal hemorrhage, and/or minor/major bleedings during surgical interventions. ABS is a standardized herbal mixture and has rapid hemostatic effect with topical application, and their ratios in the 100 mL extract are 7 g *Glycyrrhiza glabra* L. (Fabaceae), 8 g *Vitis vinifera* L. (Vitaceae), 7 g *Alpinia officinarum* Hance (Zingiberaceae), 5 g *Thymus vulgaris* L. (Lamiaceae), and 6 g *Urtica dioica* L. (Urticaceae). These herbs have also other biological effects than hemostatic actions such as antibacterial, antiinflammatory, antioxidant, and antitumoral effects [14]. In a previous study, ABS was shown to possess antioxidant and antimutagenic properties that could be of potential value in the fields of medicine and dentistry [15].

To our knowledge, this is the only study that investigates ABS's effect on epithelial dysplasia. The aim of this study is to evaluate ABS's chemopreventive effects on DMBA-induced oral epithelial dysplasia. The presence and degree of the lesions were evaluated histopathologically.

Materials and methods

Animals

Male Sprague Dawley rats (200–250 g) purchased from the animal breeding laboratory of Kobay (Ankara, Turkey) were used in the experiments. The animals were left for 1 week for acclimatization into animal room conditions and were maintained on standard pellet diet and water ad libitum. The animals were housed in polysulfone cages at 21–24 °C, 40–45% humidity, and light-controlled (12-h light/12-h dark) conditions at Laboratory Animals Breeding and Experimental Research Center, Gazi University (Ankara, Turkey). A minimum of ten rats were used in each group for biological activity studies. The

present study was performed according to the international rules considering the animal experiments and biodiversity rights (Gazi University Ethical Council Project Number—G.U.ET-12.094).

Biological activity studies

A total of 40 rats were randomized into four groups of ten rats in each group. Group 1 animals were painted with 0.5% DMBA (Sigma-Aldrich, Milwaukee, WI, USA) in liquid paraffin three times a week on the right buccal pouches for 14 weeks [16–18]. Group 2 animals were painted with DMBA three times a week, and on the alternate days of DMBA applications, ABS was applied topically on the right buccal pouches. DMBA and ABS were administered in different days to eliminate the risk of DMBA dissolving with the application of ABS. A total of 0.4 cc ABS has been used in each application. Group 3 animals received topical ABS application on the right buccal pouches three times a week for 14 weeks. Group 4 animals served as control group and treated with liquid paraffin (vehicle) alone three times a week for 14 weeks on the right buccal pouches. The experimental part of the study was terminated at the end of 14th week, and all rats were sacrificed by intraperitoneal injection of 10% ketamine and xylazine mixture at fatal dosage.

Histopathology

Systemic necropsy was performed to the sacrificed animals. For histopathological examination, right buccal mucosal tissues were fixed in 10% formalin and routinely processed and embedded with paraffin, 5-µm sections were cut in a rotary microtome and stained with hematoxylin and eosin (HE). Stained samples were examined under a light microscope (Olympus CX41 attached Kameram® Digital Image Analyze System). WHO classification of precursor and related lesions were used to evaluate the presence and the degree of dysplasia [9]. Hyperplasia, hyperkeratosis, neoplasia presence, and degrees were also scored as no lesion (–), mild (+), moderate (++) and severe (+++).

Antioxidant activity

Thiobarbituric acid reaction substances TBARS method

Ten percent of tissue homogenate was prepared by homogenization of 1 g tissue with 9 mL 0.25 M sucrose buffer. Homogenate was first centrifuged at 1000×g for 10 min and later at 2000×g for 30 min, and residue filtrate was collected. 1,1,3,3-Tetraethoxypropane as a standard procedure was used in order to construct calibration curve. Standard solutions in 800, 400, 200, 100, 50, and 25 nmol/mL concentration were prepared from 800 nmol/

mL concentration stock solution, and results were calculated given weight to age in TBARS/g term [19, 20].

Assignment of T-SH groups

The tissues were homogenized with 0.02 mol EDTA- Na_2 , and the homogenates were mixed with Tris buffer (pH 8.9) and Ellman Reagent. After adding absolute methanol to the mixture, color formation was observed 15 min later. The mixture centrifuged at $3000\times g$ for 15 min and absorbance values of supernatants were measured at 412 nm. GSH standard solution was used in the preparation of calibration curve. Standard solutions of 2, 1, 0.5, 0.25, and 0.125×10^{-4} mol concentrations were prepared from 2.10^{-4} mol concentration stock solution, and results were calculated as age weighted in $\mu\text{mol/g}$ term [20].

Assigning of GSH (NP-SH) groups

The tissues were homogenized with 0.02 mol EDTA- Na_2 , and the homogenates were mixed with 50% trichloro acetic acid and distilled water. The tubes centrifuged at $3000\times g$ for 15 min. The supernatant part was mixed with Tris buffer (pH 8.9) and Ellman Reagent. GSH standard solution was used in the preparation of calibration curve. Standard solutions of 2, 1, 0.5, 0.25, and 0.125×10^{-4} mol concentrations were prepared from 2.10^{-4} mol concentration stock solution, and results were calculated as age weighted in $\mu\text{mol/g}$ term [20].

Statistical analysis

The data obtained in this study has been analyzed with the SPSS 20 package. Test of Kruskal Wallis-H was employed for examining differences of groups not having normal distributions. Post hoc multiple comparisons tests were used to define groups with differences in cases of significant group differences resulting from Kruskal Wallis-H. Chi-Square test has been used to examine the relationships among the groups of nominal variables. When compiling the results, the level of significance was defined as 0.05, and in the case of $p < 0.05$, it

indicates a significance, while $p > 0.05$ points out an insignificant difference/relationship.

Results

The histopathological features observed in the buccal mucosa of the control group show well-defined and intact epithelial layers similar to that of ABS-alone treated group. In the DMBA-alone treated group, severe dysplasia, hyperkeratosis, and hyperplasia were noted. Epithelial neoplasia was observed in two animals at the DMBA-alone treated group. In the DMBA + ABS-treated group, intact and well-defined epithelium was observed in 50% of animals. Mild dysplasia and hyperkeratosis were observed in five animals, and mild hyperplasia was observed in four animals in the DMBA + ABS-treated group. Dysplasia, hyperplasia, and hyperkeratosis degrees observed in the DMBA-alone treated group were higher than that of other groups, and the difference between DMBA-alone treated groups and other groups were statistically significant ($p < 0.05$) (Table 1).

TBARS levels, GSH, and T-SH status were evaluated for the measurement of the lipid peroxidation in the tissue samples. TBARS levels in the DMBA + ABS-treated group were significantly lower according to the DMBA-alone treated group while GSH and T-SH status were significantly higher (Table 2).

Discussion

In the present study, we observed only mild dysplasia, hyperplasia, and hyperkeratosis in the DMBA + ABS group, while severe dysplasia, hyperplasia, and hyperkeratosis were observed in the DMBA-alone treated group (Fig. 1). This effect should be due to the antioxidant and antitumoral properties of the herbs ABS contains. It is suggested that, with the antioxidant and antitumoral effects, *T. vulgaris*, *V. vinifera*, and *G. glabra* should be inhibiting ROS production and lipid peroxidation caused by DMBA. Also, *G. glabra* should be inhibiting chronic inflammation by its antiinflammatory properties. It has been reported that *V. vinifera* has cytotoxic effects

Table 1 Scored and graded histopathologic findings according to groups

Case	Neoplasia	Dysplasia	Hyperplasia	Hyperkeratosis
Control	–	–	–	–
ABS	–	–	–	–
DMBA	–/+	+++	++/+++	++/+++
DMBA + ABS	–	–/+	–/+	–/+

These sections were scored as no lesion (–), mild (+), moderate (++) and severe (+++) for any neoplastic, dysplastic, hyperplastic, and hyperkeratotic changes

ABS, Ankaferd BloodStopper®; DMBA, 7,12-dimethylbenz[a]anthracene

Table 2 Effects of the test materials on TBARS, GSH, and T-SH levels on tissues

Test material	Statistical mean $\pm$ S.D.		
	TBARS (nmol/g)	GSH (μ mol/g)	T-SH (μ mol/g)
Control	227.4 $\pm$ 14.9	6.8 $\pm$ 2.7	8.5 $\pm$ 2.2
ABS	159.1 $\pm$ 9.3*	8.9 $\pm$ 1.4**	10.8 $\pm$ 1.4**
DMBA	234.3 $\pm$ 15.1	3.8 $\pm$ 1.1	4.7 $\pm$ 1.9
DMBA + ABS	163.7 $\pm$ 8.2*	7.4 $\pm$ 2.1*	9.5 $\pm$ 1.6*

Bold characters describes statistically difference

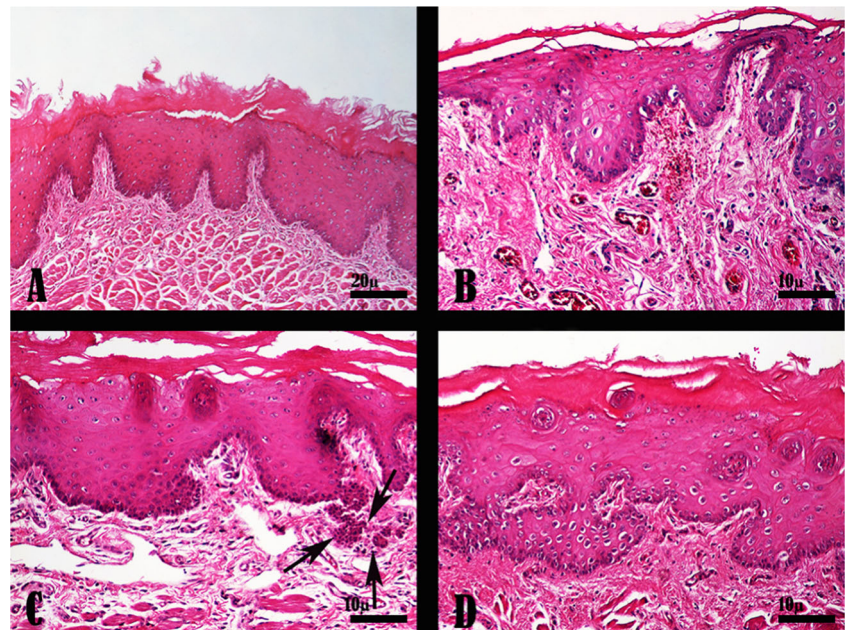
S.D., standard deviation; ABS, Ankaferd BloodStopper®; DMBA, 7,12-dimethylbenz[a]anthracene

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

against some kinds of cancer cells, and *V. vinifera* should be responsible for the prevention of neoplasia formation in this study [21]. In a previous study, *V. vinifera* shows antiproliferative effects on cancer cells and induces cytotoxic effects on cancer cells. Authors also noted that *V. vinifera* does not show cytotoxic effects on normal cells while having an antitumoral effect on cancer cells [22]. Aghbali et al. reported that *V. vinifera* extract contains potential bioactive components for the treatment of oral squamous cell carcinoma [23]. In addition, natural antioxidants can protect the cells from different diseases due to their capability of scavenging free radicals such as cancer [24]. *V. vinifera* antioxidants have an increasing popularity because of their potential anticancer effects. The anticancer effects of grape antioxidants have been demonstrated in in vitro and in vivo models [25–29]. *V. vinifera* antioxidants have been demonstrated to induce cell-cycle arrest and apoptosis in cancer cells [30]. It is also shown that they prevent carcinogenesis and cancer progression in rodent models [31, 32]. *V. vinifera* antioxidants exert potential anticancer activity by acting on multiple cellular events associated with tumor initiation, promotion, and progression because of their

variety such as resveratrol, quercetin, and catechin. It is proposed that *V. vinifera* includes antioxidant, antiinflammatory, and antiproliferative activities as potential anticancer effects [33]. It is also known that *V. vinifera* antioxidants modify estrogen receptor (ER) and therefore related to gynecological cancers such as breast cancer [34, 35]. *V. vinifera* anthocyanidins showed estrogen-inducible cell proliferation in MCF-7 breast cancer cell line but not in the receptor-negative MDA-MB-231 cell line. The receptor antagonist 4-hydroxytamoxifen can block the anthocyanidins-induced cell proliferation. The combination treatments of anthocyanidins with estradiol-reduced proliferative activity of estradiol strongly suggest that the estrogenic activity of certain grape is relevant to its beneficial activity against estrogen-dependent cancers [36]. *V. vinifera* has been reported to inhibit expression of epidermal growth factor receptor (EGFR) in head and neck squamous cell carcinoma (HNSCC) cells which also caused an inhibition of the phosphorylation of extracellular signal-regulated kinase (ERK1/2), the highly conserved Ras/mitogen-activated protein kinase (MAPK)-dependent pathway (one of EGFR major downstream pathways) [37].

Fig. 1 Histopathological view of neoplasia, dysplasia, hyperplasia, and hyperkeratosis in the test material-administered animals. All sections were stained with hematoxylin & eosin (HE), and scale bars were given on the figures. **a** Control group—mature squamous epithelium. **b** Ankaferd group—squamous epithelium histology within normal limits. **c** DMBA group—hyperkeratosis, severe dysplastic changes, and neoplastic cell invasions in the lamina propria (arrows). **d** DMBA and Ankaferd-treated group—mild hyperkeratosis and dysplastic changes



Conclusions

In conclusion, the results of this study suggest that Ankaferd Bloodstopper® has chemopreventive effects against DMBA-induced oral epithelial dysplasia. Further studies are however needed to clarify the exact molecular mechanism for the chemopreventive potential of ABS in DMBA-induced oral epithelial dysplasia.

Funding This study was funded by the Scientific Research Project of Gazi University (Grant No. 02-2013-01).

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval All procedures performed in studies involving animals were in accordance with the ethical standards of the institution or practice at which the studies were conducted (Gazi University Ethical Council Project Number—G.U.ET-12.094). This article does not contain any studies with human participants performed by any of the authors.

Informed consent Informed consent was obtained from all individual participants included in the study.

References

1. Yılmaz M (1999) DMBA (dimetilbenzantrasen) ile dilde meydana getirilen kimyasal kanser modeli üzerinde karsinogenezis sürecinin histopatolojik takibi ve selenyumun karsinogenezis sürecine etkisinin incelenmesi. Uzmanlık Tezi, Ankara
2. Stewart BW, Christopher PW (2014) World Cancer Report 2014. WHO Press
3. Villa A, Villa C, Abati S (2011) Oral cancer and oral erythroplakia: an update and implication for clinicians. Aust Dent J 56(3):253–256. <https://doi.org/10.1111/j.1834-7819.2011.01337.x>
4. Jemal A, Bray F, Center MM, Ferlay J, Ward E, Forman D (2011) Global cancer statistics. CA Cancer J Clin 61(2):69–90
5. Siegel RL, Miller KD, Jemal A (2018) Cancer statistics, 2018. CA Cancer J Clin 68:7–30
6. Petersen PE (2009) Oral cancer prevention and control the approach of the World Health Organization. Oral Oncol 45(4–5):454–460. <https://doi.org/10.1016/j.oraloncology.2008.05.023>
7. Ferlay J, Shin HR, Bray F, Forman D, Mathers C, Parkin DM (2010) Estimates of worldwide burden of cancer in 2008: GLOBOCAN 2008. Int J Cancer 127(12):2893–2917. <https://doi.org/10.1002/ijc.25516>
8. Yüçetaş Ş (2005) Ağız ve Çevre Dokusu Hastalıkları. Atlas Kitapçılık
9. Barnes L, Eveson JW, Reichart P, Sidransky D (2005) World Health Organisation classification of tumors. IARC Press, Lyon
10. Izumo T (2011) Oral premalignant lesions: from the pathological viewpoint. Int J Clin Oncol 16(1):15–26. <https://doi.org/10.1007/s10147-010-0169-z>
11. Manoharan S, Vasanthaselman M, Silvan S, Baskaran N, Kumar Singh A, Vinoth Kumar V (2010) Carnosic acid: a potent chemopreventive agent against oral carcinogenesis. Chem Biol Interact 188(3):616–622. <https://doi.org/10.1016/j.cbi.2010.08.009>
12. Burki HR, Okita GT (1969) Effect of oral copper sulfate on 7,12-dimethylbenz(alpha)anthracene carcinogenesis in mice. Br J Cancer 23(3):591–596
13. Salley JJ (1954) Experimental carcinogenesis in the cheek pouch of the Syrian hamster. J Dent Res 33(2):253–262
14. Fırat HC, Özdemir O, Koşar A, Göker H, Haznedaroğlu İC (2009) Ankaferd Blood Stopper Annual Review of Arkaferd 08-09. Naviga Scientific Publication
15. Ugur A, Sarac N, Cankal DA, Ozle M (2016) The antioxidant and antimutagenic activities of Ankaferd blood stopper, a natural hemostatic agent used in dentistry. Turk J Med Sci 46:657–663
16. Dhanarasu S, Selvam M, Salama SM, Shanmugam M, Sethuraman P (2010) *Terminalia arjuna* (Roxb.) modulates circulatory antioxidants on 7,12-dimethylbenz(a)anthracene-induced hamster buccal pouch carcinogenesis. Oman Med J 25(4):276–281. <https://doi.org/10.5001/omj.2010.81>
17. Manoharan S, Sindhu G, Vinothkumar V, Kowsalya R (2012) Berberine prevents 7,12-dimethylbenz(a)anthracene-induced hamster buccal pouch carcinogenesis: A biochemical approach. Eur J Cancer Prev 21:182–192
18. Panjamurthy K, Manoharan S, Nirmal MR, Vellaichamy L (2009) Protective role of Withaferin-A on immunorepression of p53 and bcl-2 in 7,12-dimethylbenz(a)anthracene-induced experimental oral carcinogenesis. Investig New Drugs 27(5):447–452. <https://doi.org/10.1007/s10637-008-9199-z>
19. Kuranel E (2012) *Plantago lanceolata* Bitkisinin Yara İyileştirici Özelliklerinin Araştırılması. Yüksek Lisans, İnönü Üniversitesi, Malatya
20. Sedlak J, Lindsay RH (1968) Estimation of total, protein-bound, and nonprotein sulfhydryl groups in tissue with Ellman's reagent. Anal Biochem 25(1):192–205
21. Ye X, Krohn RL, Liu W, Joshi SS, Kuszynski CA, McGinn TR, Bagchi M, Preuss HG, Stohs SJ, Bagchi D (1999) The cytotoxic effects of a novel IH636 grape seed proanthocyanidin extract on cultured human cancer cells. Mol Cell Biochem 196(1–2):99–108
22. Giovannelli L, Innocenti M, Santamaria AR, Bigagli E, Pasqua G, Mulinacci N (2014) Antitumoural activity of viniferin-enriched extracts from *Vitis vinifera* L. cell cultures. Nat Prod Rep 28(22):2006–2016. <https://doi.org/10.1080/14786419.2014.924935>
23. Aghbali A, Hosseini SV, Delazar A, Gharavi NK, Shahneh FZ, Orangi M, Bandehagh A, Baradaran B (2013) Induction of apoptosis by grape seed extract (*Vitis vinifera*) in oral squamous cell carcinoma. Bosn J Basic Med Sci 13(3):186–191
24. Niture SK, Rao US, Srivenugopal KS (2006) Chemopreventative strategies targeting the MGMT repair protein: Augmented expression in human lymphocytes and tumor cells by ethanolic and aqueous extracts of several Indian medicinal plants. Int J Oncol 29(5):1269–1278
25. Clifford AJ, Ebeler SE, Ebeler JD, Bills ND, Hinrichs SH, Teissedre PL, Waterhouse AL (1996) Delayed tumor onset in transgenic mice fed an amino acid-based diet supplemented with red wine solids. Am J Clin Nutr 64(5):748–756
26. Jung KJ, Wallig MA, Singletary KW (2006) Purple grape juice inhibits 7,12-dimethylbenz(a)anthracene (DMBA)-induced rat mammary tumorigenesis and in vivo DMBA-DNA adduct formation. Cancer Lett 233(2):279–288. <https://doi.org/10.1016/j.canlet.2005.03.020>
27. Kim H, Hall P, Smith M, Kirk M, Prasain JK, Barnes S, Grubbs C (2004) Chemoprevention by grape seed extract and genistein in carcinogen-induced mammary cancer in rats is diet dependent. J Nutr 134(12):3445–3452
28. Morre DM, Morre DJ (2006) Anticancer activity of grape and grape skin extracts alone and combined with green tea infusions. Cancer Lett 238(2):202–209. <https://doi.org/10.1016/j.canlet.2005.07.011>
29. Soleas GJ, Grass L, Josephy PD, Goldberg DM, Diamandis EP (2002) A comparison of the anticarcinogenic properties of four red wine polyphenols. Clin Biochem 35(2):119–124

30. Aggarwal BB, Bhardwaj A, Aggarwal RS, Seeram NP, Shishodia S, Takada Y (2004) Role of resveratrol in prevention and therapy of cancer: Preclinical and clinical studies. *Anticancer Res* 24(5A):2783–2840
31. Ebeler SE, Brennen CA, Kim GS, Jewell WT, Webb MR, Chacon-Rodriguez L, MacDonald EA, Cramer AC, Levi A, Ebeler JD, Islas-Trejo A, Kraus A, Hinrichs SH, Clifford AJ (2002) Dietary catechin delays tumor onset in a transgenic mouse model. *Am J Clin Nutr* 76(4):865–872
32. Garvin S, Ollinger K, Dabrosin C (2006) Resveratrol induces apoptosis and inhibits angiogenesis in human breast cancer xenografts *in vivo*. *Cancer Lett* 231(1):113–122. <https://doi.org/10.1016/j.canlet.2005.01.031>
33. Seeram NP, Adams LS, Henning SM, Niu Y, Zhang Y, Nair MG, Heber D (2005) *In vitro* antiproliferative, apoptotic and antioxidant activities of punicalagin, ellagic acid and a total pomegranate tannin extract are enhanced in combination with other polyphenols as found in pomegranate juice. *J Nutr Biochem* 16(6):360–367
34. Klinge CM, Risinger KE, Watts MB, Beck V, Eder R, Jungbauer A (2003) Estrogenic activity in white and red wine extracts. *J Agric Food Chem* 51(7):1850–1857. <https://doi.org/10.1021/jf0259821>
35. Schlachterman A, Valle F, Wall KM, Azios NG, Castillo L, Morell L, Washington AV, Cubano LA, Dharmawardhane SF (2008) Combined resveratrol, quercetin, and catechin treatment reduces breast tumor growth in a nude mouse model. *Transl Oncol* 1(1):19–27
36. Schmitt E, Stopper H (2001) Estrogenic activity of naturally occurring anthocyanidins. *Nutr Cancer* 41(1–2):145–149. <https://doi.org/10.1080/01635581.2001.9680625>
37. Sun Q, Prasad R, Rosenthal E, Katiyar SK (2011) Grape seed proanthocyanidins inhibit the invasive potential of head and neck cutaneous squamous cell carcinoma cells by targeting EGFR expression and epithelial-to-mesenchymal transition. *BMC Complement Altern Med* 11:134. <https://doi.org/10.1186/1472-6882-11-134>